

activation values of neurons, and b is the bias term. All neurons in a feature map share the same weights (kernel) and bias term, but different activation values that correspond to the values of the receptive field corresponding to that neuron in the feature map. This is one of the reasons why computations are less in a CNN compared to a DNN (deep neural network). A feature learnt by a neuron at one particular location will allow it to detect the same feature anywhere else in the input, regardless of the location. This is another important feature of the CNN as compared to a DNN, which is not translational invariant.

Next, consider the convolution layers after the first convolution layer. The neuron at location (x, y) in the feature map k in a convolutional layer l gets its input from all the neurons in the previous layer $l-1$, which are located in rows:

$x * sw \text{ to } * sh + kw - 1$, columns $y * sh \text{ to } y * sw + kh - 1$

...and for all feature maps in the layer $l-1$. These are the same inputs to all neurons located in the same row x and column y , in all the feature maps. This explains how the features learnt in one layer get integrated to learn combined features or complex patterns in successive layers. This also explains how a CNN learns distinct features at the initial layers; and convolutes to integrate basic features to learn complex distinct features or patterns in the intermediate convolution layers, before finally learning objects present in the input in the last convolution layers. As discussed earlier, any feature, pattern or object learnt by the CNN at any layer will allow it to detect this anywhere else in the input, independent of the location. So one needs to be extra careful when using a CNN to detect combined objects that are similar in the input. For example, in the case of face detection, if one is interested in

studying the features of the eye and eyebrows together as a combination, then normal CNN would learn the features of the eyebrow and the eye as two different features, and it may not help in learning the minor differences of the combination of features. In that case, the convolution layers have to be tweaked to learn those features appropriately.

Activation feature maps in CNN

Although activations are generated for each neuron, they may not be propagated to the final prediction.

Let us now get the equations to determine the output of a neuron at a convolutional layer. For a grey scale image (single channel) the output of the neuron in the first convolution layer located at (x, y) of the feature map k is given by the following:

$zx, y = \sum_{m=1 \text{ to } kh} (\sum_{n=1 \text{ to } kw} (ai, j * wm, n)) + bk \text{ for } i = m * sh + kh - 1 \text{ and } j = n * sw - 1 \text{ and } ai, j$

...which will be the pixel values of the input image. And by:

$kh, kw, sh, \text{ and } sw$

...which are the height and width of the kernel and the horizontal and vertical strides, respectively.

The output of the neuron in any convolutional layer l (l is not the first convolutional layer) and for the feature map k is given by the following:

$zx, y, k = \sum_{m=1 \text{ to } kh} (\sum_{n=1 \text{ to } kw} (\sum_{fm=1 \text{ to } kl-1} (ai, j, fm * wm, n, fm, k))) + bk \text{ for } i = m * sh + kh - 1 \text{ and } j = n * sw - 1 \text{ and } ai, j, fm$

...which will be the activation values of previous layer $l-1$. And by:

$kh, kw, sh, \text{ and } sw$

...which are the height and width of the kernel, and the horizontal and

vertical strides, respectively.

zx, y, k

The above is the output of the neuron located at row m , column n in the feature map k of the convolutional layer l .

$kl-1$

The above is the number of feature maps in the layer $l-1$ (previous layer).

ai, j, fm

The above is the activation output of the neuron located in layer $l-1$, row i , column j , and feature map fm . bk is the bias term for feature map k (in layer l).

wi, j, fm, k

The above is the weight (kernel) of any neuron in the feature map k of the layer l and its input located at row m , column n , relative to the neuron's receptive field and feature map fm .

The feature maps produced in the convolution layer are input to the next layer (pooling layer) after applying the activation function. The feature map generated after applying the activation function is called the activation feature map. The visualisation of the activation feature map for a filter in a particular convolution layer depicts the regions of the input that are activated in the feature map after applying the filter. In each activation feature map, the neurons have a range of activation values with the maximum value representing the most activated neuron, the minimum value representing the least activated neuron, and the value zero representing a neuron not activated.

Let us consider an illustration. The AlexNet model is used for transfer learning to build a binary classifier that classifies the given flight image to either a passenger flight or a fighter flight. AlexNet has five convolutional